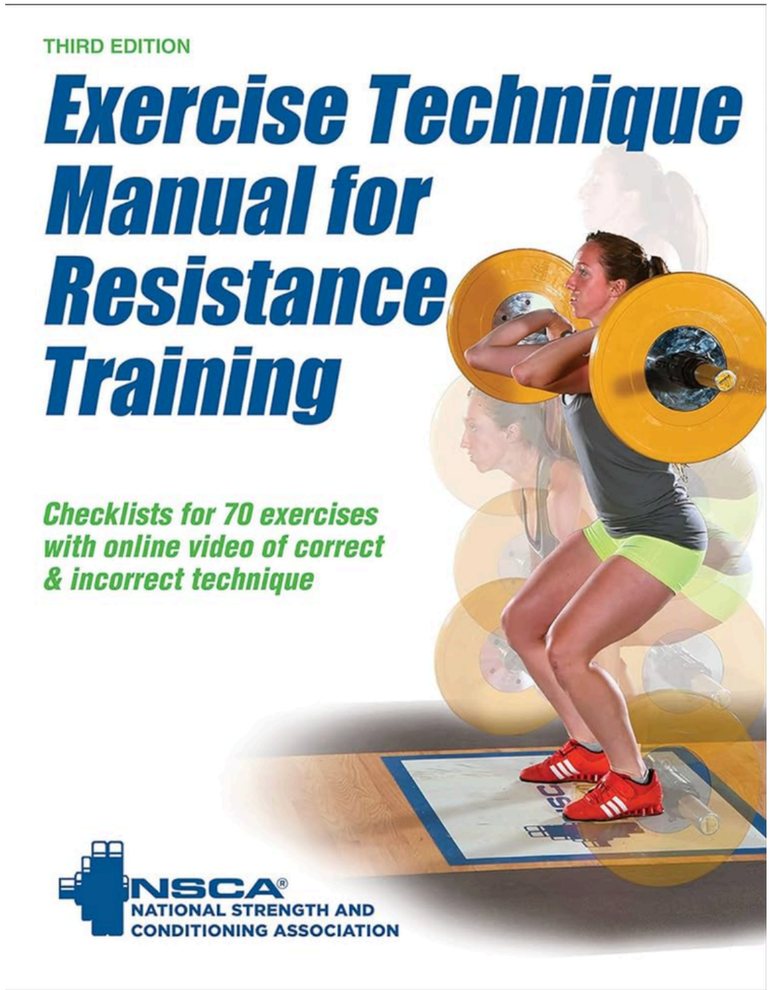
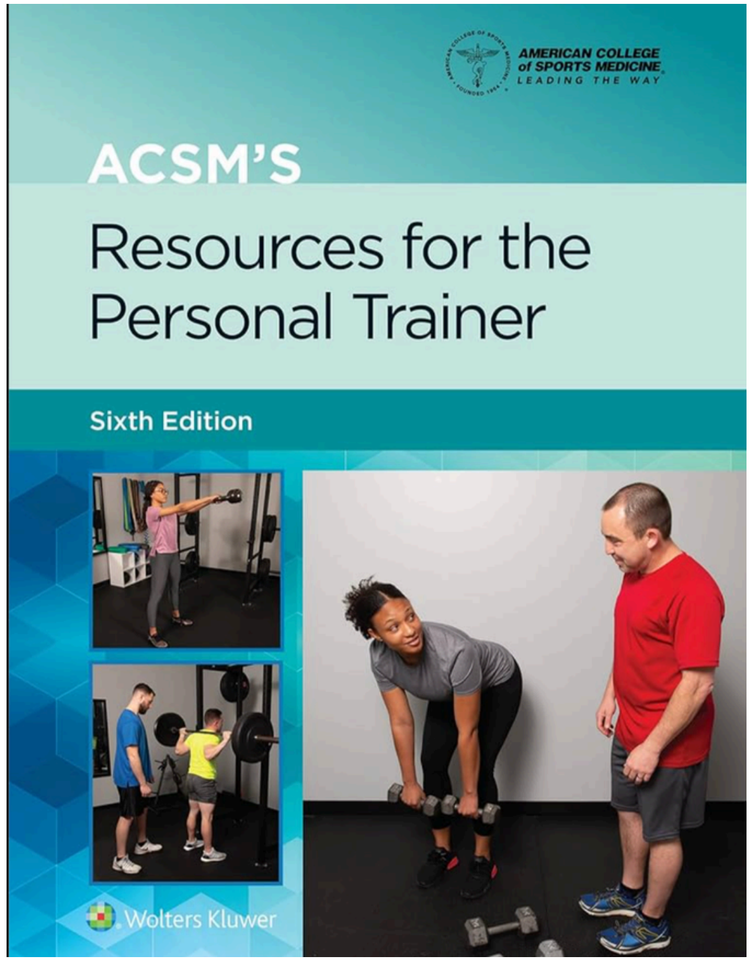
Week 4





36 question answer pairs

```python
from langchain_text_splitters import RecursiveCharacterTextSplitter

CHUNK_SIZE = 500
CHUNK_OVERLAP = 50
RETRIEVAL_K = 2


def base_chunking(docs_list):
    text_splitter = RecursiveCharacterTextSplitter(
        chunk_size=CHUNK_SIZE,
        chunk_overlap=CHUNK_OVERLAP,
        add_start_index=True,
    )
    doc_splits = text_splitter.split_documents(docs_list)
    print(f"  Split into {len(doc_splits)} chunks")
    return doc_splits
```

Recursively split based on the following order:

“\n\n”

“\n”

“ ”

“”

baseline chunking

%99 retrieval relevance accuracy?

Example - Not really the correct context

Rag Evaluation Results Website

RETRIEVAL_RELEVANCE_INSTRUCTIONS = """You are a teacher grading a quiz. You will be given a QUESTION and a set of FACTS provided by the student. Here is the grade criteria to follow:
(1) Your goal is to identify FACTS that are completely unrelated to the QUESTION
(2) If the facts contain ANY keywords or semantic meaning related to the question, consider them relevant
(3) It is OK if the facts have SOME information that is unrelated to the question as long as (2) is met

problematic

You are an expert evaluator assessing whether retrieved context contains sufficient information to answer a given question.

Your task is to determine if the CONTEXT provides the specific information needed to answer the QUESTION.

Follow these rules strictly:
- Do NOT rely on your own knowledge. Use ONLY what is explicitly written in the CONTEXT.
- Semantic similarity or shared keywords are NOT enough. The CONTEXT must contain actual information that directly addresses what the QUESTION is asking.
- If the QUESTION asks for a definition, the CONTEXT must contain that definition.
- If the QUESTION asks for a list (e.g. "what are the three phases"), the CONTEXT must explicitly mention those items.
- If the QUESTION has multiple parts, the CONTEXT must address ALL parts to be considered relevant.

Evaluation criteria:
- True: The CONTEXT contains specific information that directly answers the QUESTION (or all parts of it). A person reading only the CONTEXT could reasonably answer the QUESTION.
- False: The CONTEXT is on a related topic but does not contain the specific information needed to answer the QUESTION, OR only addresses some parts of a multi-part question.

Here are three examples:

evaluators.py

retrieval relevance down to %80

Some chunks fall short to fully anwer and sometimes the two retrieved chunks are consecutive => increase chunk size

<u>short chunk</u>

chunk size = 1500

overlap = 150

Shoulder abduction vs adduction

Some chunk still fall short, or more retrievel is required

Retrived context is still relevant in terms of semantics and keywords, but doesnt really answer the question

increase k?

Is system prompt too strict?

k=16, nearly perfect

{"explanation":"The context discusses the seated dumbbell shoulder press and highlights that performing the exercise while seated is stricter than standing, as it prevents cheating the weight upward using momentum generated by the legs. However, it does not explicitly state whether one should perform it seated or standing; it only states the seated position is preferable for control. Since the context does not give a definitive recommendation between the two options, it does not adequately answer the question.","relevant":false}

Semantic Chunker

Split documents into sentences by regex

Measure similarity of embeddings of sentences

Split when similarity change is high

Percentile 90 gives nearly 10k chunks

Max chunk size is 5k, average is 550

Results

Why does evaluator fall into this? (question id 4)

evaluator is gpt4o, 128k context, we provide roughly 10k

All Experiments

MarkdownHeaderTextSplitter

The author already did the logical partitioning, use it

<u>Agentic Splitting</u> - Experimental method of splitting text with an agent-like system. Good for if you believe that token cost will trend to $0.00

Good when you have a dense document full of “hap bilgi”

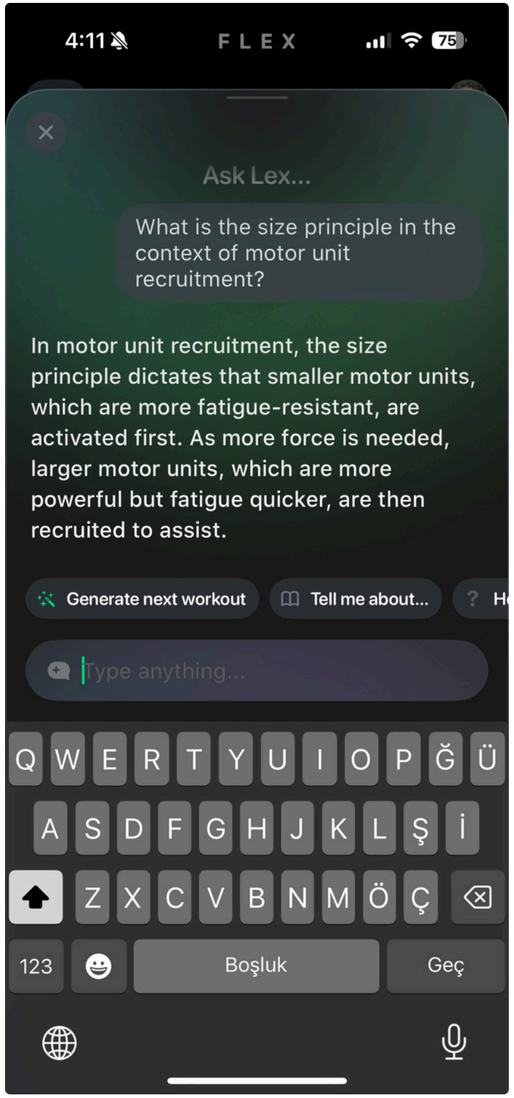
https://github.com/FullStackRetrieval-com/RetrievalTutorials/blob/a4570f3c4883eb9b835b0ee18990e62298f518ef/tutorials/LevelsOfTextSplitting/5_Levels_Of_Text_Splitting.ipynb

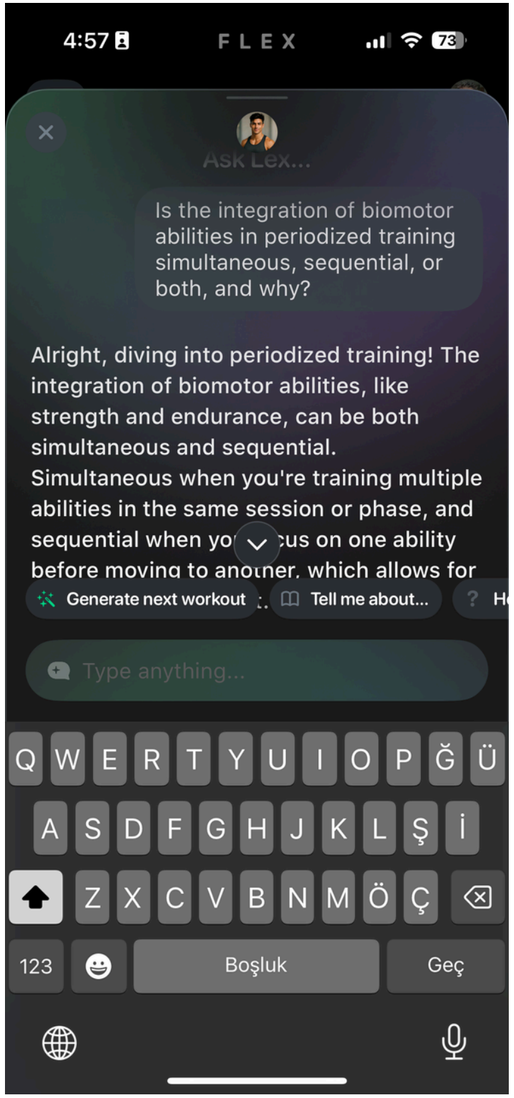
Hierarchical Chunking

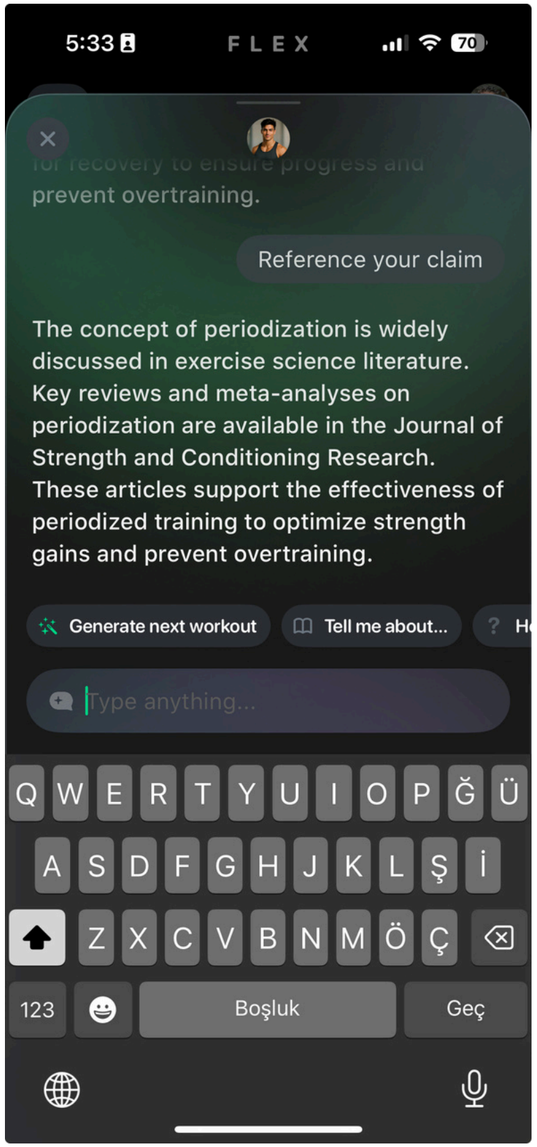
```
parent_splitter = SemanticChunker(
    embeddings=embeddings,
    breakpoint_threshold_type="percentile",
    breakpoint_threshold_amount=95,
    buffer_size=1,
)
```

```
retriever = ParentDocumentRetriever(
    vectorstore=vectorstore,
    docstore=docstore,
    child_splitter=child_splitter,
    search_kwargs={"k": search_k},
)
```

What about the other metrics??


4:11
FLEX
Ask Lex...
What is the size principle in the context of motor unit recruitment?
In motor unit recruitment, the size principle dictates that smaller motor units, which are more fatigue-resistant, are activated first. As more force is needed, larger motor units, which are more powerful but fatigue quicker, are then recruited to assist.
Generate next workout
Tell me about...
Type anything...
Boşluk
Geç


4:57
FLEX
Ask Lex...
Is the integration of biomotor abilities in periodized training simultaneous, sequential, or both, and why?
Alright, diving into periodized training! The integration of biomotor abilities, like strength and endurance, can be both simultaneous and sequential.
Simultaneous when you're training multiple abilities in the same session or phase, and
Generate next workout
Tell me about...
Type anything...
Boşluk
Geç



Distillation!

Hi there,

We're happy to share that we've automatically moved your organization from **Usage Tier 1** to **Usage Tier 2** based on your usage history on our platform. For most organizations, this comes with an increase in rate limits across several models. You can review your new usage tier and limits in your account settings.

For a complete overview of how limits and usage tiers work, please refer to our documentation.

Thank you,
The OpenAI team

Next??

Play with embedding model?
llm-craft query?
re-retrive?